Chromosoma (2003) 112:152–158
DOI 10.1007/s00412-003-0255-3

RESEARCH ARTICLE

Jiři Macas · Alice Navrátilová · Tibor Mészáros

Sequence subfamilies of satellite repeats related to rDNA intergenic spacer are differentially amplified on *Vicia sativa* chromosomes

Received: 14 March 2003 / Accepted: 6 August 2003 / Published online: 27 September 2003
© Springer-Verlag 2003

Abstract We cloned and sequenced the *Vicia sativa* 25S-18S rDNA intergenic spacer (IGS) and the satellite repeat S12, thought to be related to the spacer sequence. The spacer was shown to contain three types of subrepeats (A, B, and C) with monomers of 173 bp (A), 10 bp (B), and 66 bp (C), separated by unique or partially duplicated sequences. Two spacer variants were detected in *V. sativa* that differed in length (2990 and 3168 bp) owing to an extra copy of the subrepeat A. The A subrepeats were also shown to be highly homologous to the satellite repeat S12, which is located in large clusters on chromosomes 4, 5, and 6, and is not associated with the rDNA loci. Sequencing of additional S12 clones retrieved from a shotgun genomic library allowed definition of three subfamilies of this repeat based on minor differences in their nucleotide sequences. Two of these subfamilies could be discriminated from the rest of the S12 sequences as well as from the IGS A subrepeats using specific oligonucleotide primers that labeled only a subset of the S12 loci when used in the primed in situ DNA labeling (PRINS) reaction on mitotic chromosomes. These experiments showed that, in spite of the high overall similarity of the IGS A subrepeats and the S12 satellite repeats, there are S12 subfamilies that are divergent from the common consensus and are present at only some of the chromosomes containing the S12 loci. Thus, the subfamilies may have evolved at these loci following the spreading of the A subrepeats from the IGS to genomic regions outside the rDNA clusters.

Communicated by B. McKee

Accession numbers: GenBank AY234364–AY234374. The monomer sequences and additional information about the family of IGS-like repeat S12 will also appear in the PlantSat database (Macas et al. 2002, http://w3lamc.umbr.cas.cz/PlantSat/) under Accession name Vicia_sativa_IGS-like

J. Macas (✉) · A. Navrátilová · T. Mészáros
Institute of Plant Molecular Biology,
Laboratory of Molecular Cytogenetics,
Branišovská 31, České Budějovice, 37005 Czech Republic
e-mail: macas@umbr.cas.cz

Electronic Supplementary Material Supplementary material is available in the online version of this article at http://dx.doi.org/10.1007/s00412-003-0255-3

Introduction

In higher plants, the genes coding for the 18S, 5.8S and 25S rRNAs are clustered and transcribed as one unit. The clusters are present in multiple copies in a genome that are arranged in tandem and separated by intergenic spacers (IGSs). The spacer sequence is located between the 3′ end of the 25S gene and the 5′ end of the 18S gene of the following unit, and is from 1 kb to over 12 kb long (Rogers and Bendich 1987). The spacers contain regulatory and functional regions required for transcription of rRNA genes (Piller et al. 1990; Ueki et al. 1992; Suzuki et al. 1996), and origins of replication (Hernandez et al. 1993). They have also been shown to promote homologous recombination and amplification of adjacent sequence regions (Borisjuk et al. 2000; Urawa et al. 2001). In contrast to the rRNA genes, the spacer sequences are not conserved between species and exhibit various degrees of homology. A typical feature of plant IGS sequences is their complex structure, resulting from the presence of up to eight different tandem subrepeats with monomers of various lengths (Kato et al. 1985; Rogers and Bendich 1987; Kato et al. 1990; Ueki et al. 1992; Suzuki et al. 1996). Some of these subrepeats may carry promoter or potential transcription initiation sequences (Takaiwa et al. 1990; Suzuki et al. 1996). Moreover, termination of transcription was mapped to IGS subrepeats adjacent to the 3′ end of the 25S gene in *Vigna radiata* (Schiebel et al. 1989).

An interesting finding was reported for the legumes *Vicia faba*, *Vigna radiata* and *Phaseolus vulgaris*, which were shown to contain highly amplified satellite repeats with sequence homology to the IGS subrepeats. It has been established in all three species that these satellite sequences occur in blocks independent of the rRNA gene clusters (Maggini et al. 1991; Unfried et al. 1991; Falquet

et al. 1997). Supposing that the repeats originated from the IGS sequences, this phenomenon may represent one of the mechanisms of origin and amplification of satellite DNA in plants. On the other hand, there is also a possibility that the satellite sequences have transposed into rDNA spacers from other genomic regions, as proposed for *V. faba* (Maggini et al. 1991). In any case, more information about these repeats and their genomic organization is needed to elucidate their evolutionary dynamics in the genome.

Recently, we have cloned from common vetch (*Vicia sativa* L.) yet another satellite repeat presumably related to IGS. This sequence, designated S12, is amplified in *V. sativa* but not in other *Vicia* species tested (Nouzová et al. 2001). It has significant homology to IGS subrepeats from *Vicia angustifolia* (Ueki et al. 1992); however, its relation to the IGS sequences from *V. sativa* could not be determined owing to unavailability of sequence data from this species. Thus, in the present work we aimed to clone and sequence the *V. sativa* IGS, and to characterize IGS-like satellite repeats with respect to their sequence composition, variability, distribution on chromosomes, and possible relation to the IGS sequences.

Materials and methods

Plant material and DNA isolation

Seeds of *V. sativa* var. Ebena were obtained from the plant breeding station at Horní Moštěnice, Czech Republic. Total genomic DNA was isolated from young leaves as described by Dellaporta et al. (1983).

Cloning procedures and sequence analysis

Polymerase chain reaction (PCR) amplification of spacer sequences was performed using primers rDNA-3 (5′ GCC ACG ATC CAC TGA GAT TC 3′), and rDNA-5 (5′ ATG ACT ACT GGC AGG ATC AAC C 3′), and 20 ng of *V. sativa* genomic DNA as a template. The reaction mix (50 μl) consisted of 1×PCR buffer, 1.5 mM $MgCl_2$, 0.2 mM dNTPs, 0.2 μM primers, and 2.5 U of Platinum Taq polymerase (Invitrogen). The reaction involved 30 cycles of 94°C for 1 min, 55°C for 1 min, and 72°C for 4 min, preceded by an initial denaturation (94°C, 2 min) and followed by a final extension step (72°C, 10 min). The reaction products were gel-purified and cloned using the TOPO TA Cloning Kit (Invitrogen). Templates for sequencing were generated using the GeneJumper Kit (Invitrogen).

The shotgun genomic DNA library was prepared from *V. sativa* DNA subjected to sonication, mung bean nuclease treatment and size fractionation on an agarose gel. The fragments of about 800 bp were purified from the gel, treated with T4 DNA polymerase in the presence of dNTPs, and cloned into *Sma*I-cut and dephosphorylated plasmid vector (pBluescript II SK+). The library was screened by colony hybridization with the S12 clone (Nouzová et al. 2001) as a probe, using the AlkPhos Direct hybridization and detection kit (Amersham).

Sequencing was performed by the dideoxy chain termination method (Sanger et al. 1977). The resultant sequences were assembled in the Staden Sequence Analysis Package (Staden 1996) and further analyzed using Dotter (Sonnhammer and Durbin 1995), Tandem Repeats Finder (Benson 1999), BLAST (Altschul et al. 1997), FASTA (Pearson and Lipman 1988), and the software tools (ClustalW, BOXSHADE, MVIEW) implemented at the Biology WorkBench website (http://workbench.sdsc.edu/).

Primed in situ DNA labeling

Primed in situ DNA labeling (PRINS) reactions were performed on squash preparations of mitotic chromosomes as described by Macas et al. (2000), with the following modifications. The reaction mix (65 μl) contained 1×PCR buffer, 4 mM $MgCl_2$, 0.1 mM each of dATP, dCTP, dGTP, 0.017 mM dTTP, 0.008 mM Alexa-fluor-488-dUTP (Molecular Probes), 0.2 μM primer, and 5.2 U of Taq polymerase (Promega). Only a single primer was used in each reaction and the temperature profile consisted of DNA denaturation at 94°C for 3 min, primer annealing at 55°C for 5 min, and primer extension at 72°C for 30 min (the heating rate between annealing and extension steps was 0.1°C per second). The primers S12-4 (5′ TCT GTT TCC GTC CTA GAA AAT C 3’), S12-123 (5’ TCC GTT CCC GTC CGT GTC GTG C 3′), and S12-2 (5′ CCC GAC ACC GTC GTG CAC GTG CAC 3′) were designed according to the multiple sequence alignment of monomer sequences of S12 repeats (Fig. 2). The chromosomes were counterstained with 4′,6-diamino-2-phenylindole and observed using a Nikon Eclipse-600 epifluorescence microscope equipped with a CCD camera. The signals were collected separately using appropriate filter sets and LUCIA software (Laboratory Imaging).

Results

Cloning and analysis of *V. sativa* rDNA IGS

A PCR using primers derived from conserved plant rRNA sequences flanking the IGS was employed to amplify IGS regions from *V. sativa*. The reaction yielded two products of about 3 kb that differed slightly in size (data not shown). Cloning and sequencing of these fragments showed that they were 3059 bp (clone VS-IGS-3, Accession number AY234366) and 3237 bp (clone VS-IGS-5, Accession number AY234364) and contained almost identical sequences. Sequence analysis revealed the presence of three types of tandem subrepeats, designated A, B, and C (Fig. 1A, B). The A subrepeats have monomers of 172–173 bp and are present in two full-length and several truncated copies in the region next to the 3′ end of the 25S rRNA gene. This region has a complex structure that probably originated from a partial duplication of a sequence composed of the A subrepeats and some unique spacer sequences (Fig. 1B). An additional duplication of the first copy of the A subrepeat was also identified as a cause of the size difference between the two clones analyzed. In contrast to the A subrepeats the other two subrepeats are arranged in continuous blocks of basic repeated units of 10 bp (B subrepeats) and 66 bp (C subrepeats), respectively. A putative transcription initiation site was identified between these blocks (Fig. 1A).

Comparison with other plants revealed significant homology of *V. sativa* spacers to the IGS sequence of *V. angustifolia* ssp. *segetalis* (Ueki et al. 1992). However, despite the overall similarity, the sequences differed considerably in the number of repetitions of the spacer subrepeats, which were more amplified in the *V. sativa*

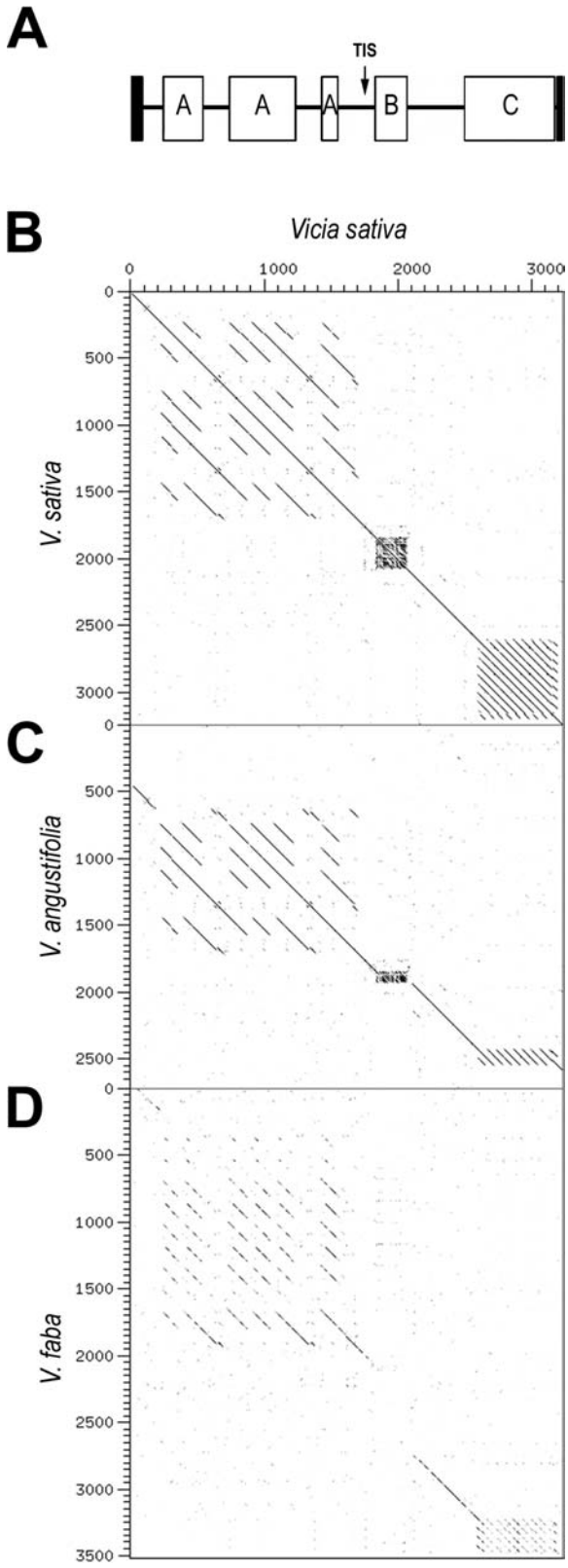


Fig. 1A–D Structure of the *Vicia sativa* intergenic spacer (IGS) and comparison with that of other species. **A** Schematic representation of clone VS-IGS-5 showing the positions of the 25S and 18S rRNA genes (*black rectangles*), regions containing subrepeats A, B and C (*open rectangles*), and putative transcription initiation site (*TIS*) identified at base 1775 by its homology with other plant TIS sequences. **B–D** A dot-matrix comparison of *V. sativa* IGS (clone VS-IGS-5, the scale corresponds to the scheme above) with itself (**B**) and with spacer sequences from *Vicia angustifolia* (Ueki et al. 1992) (**C**) and *Vicia faba* (Kato et al. 1990) (**D**). Homologous sequence regions are represented with *dots* or *lines*

IGS (Fig. 1C). Homology was also detected with spacer sequences of *V. faba* in the regions corresponding to unique sequences and to the A and C subrepeats (Fig. 1D). However, other species including legumes displayed increasing divergence of their spacer sequences (not shown).

Relation of the IGS subrepeats to the S12 satellite repeat

Sequence comparison revealed 83%–84% homology of the A subrepeats with the previously isolated clone S12 (Nouzová et al. 2001) and therefore confirmed the supposed similarity of the spacer sequences with the independent satellite amplified on several *V. sativa* chromosomes. In order to obtain more sequences for comparative analysis, a shotgun library was constructed from *V. sativa* genomic DNA and screened using the clone S12 as a probe. Nineteen positive clones were picked at random and sequenced. Five of these clones contained IGS regions including A subrepeats and the rest was composed exclusively of tandemly repeated sequences homologous to S12. Forty-six full-length monomer sequences were extracted from these clones and used for multiple sequence alignment together with A subrepeats from clones VS-IGS-3 and VS-IGS-5. The monomers displayed high homology over their entire length except for a region close to their 3′ end that allowed definition of four distinct groups of sequences (Fig. 2 and Electronic Supplementary Material). These groups comprised 6–19 monomers with mutual homology of 85%–93% (group 1), 89%–96% (group 2), 93%–100% (group 3), and 89%–98% (group 4), respectively. The homology between consensus sequences calculated for each group ranged from 87% to 96%. Monomer sequences from individual clones were always assigned to the same group, which reflects their high homogenization within the clones. Moreover, all IGS A subrepeats formed a distinct group (group 3, Fig. 2) with a consensus sequence of 173 bp and the highest average homology between the monomers (96%; see Electronic Supplementary Material for the alignment data). Two other groups (1 and 4) had a consensus of a similar size (173–174 bp), whereas the group 2 sequences were longer owing to a duplication of the 6 bp motif TGCACG (Fig. 2).

Distribution of the IGS-like repeat subfamilies on *V. sativa* chromosomes

The localization of IGS-like repeats on mitotic chromosomes was investigated using PRINS with specific

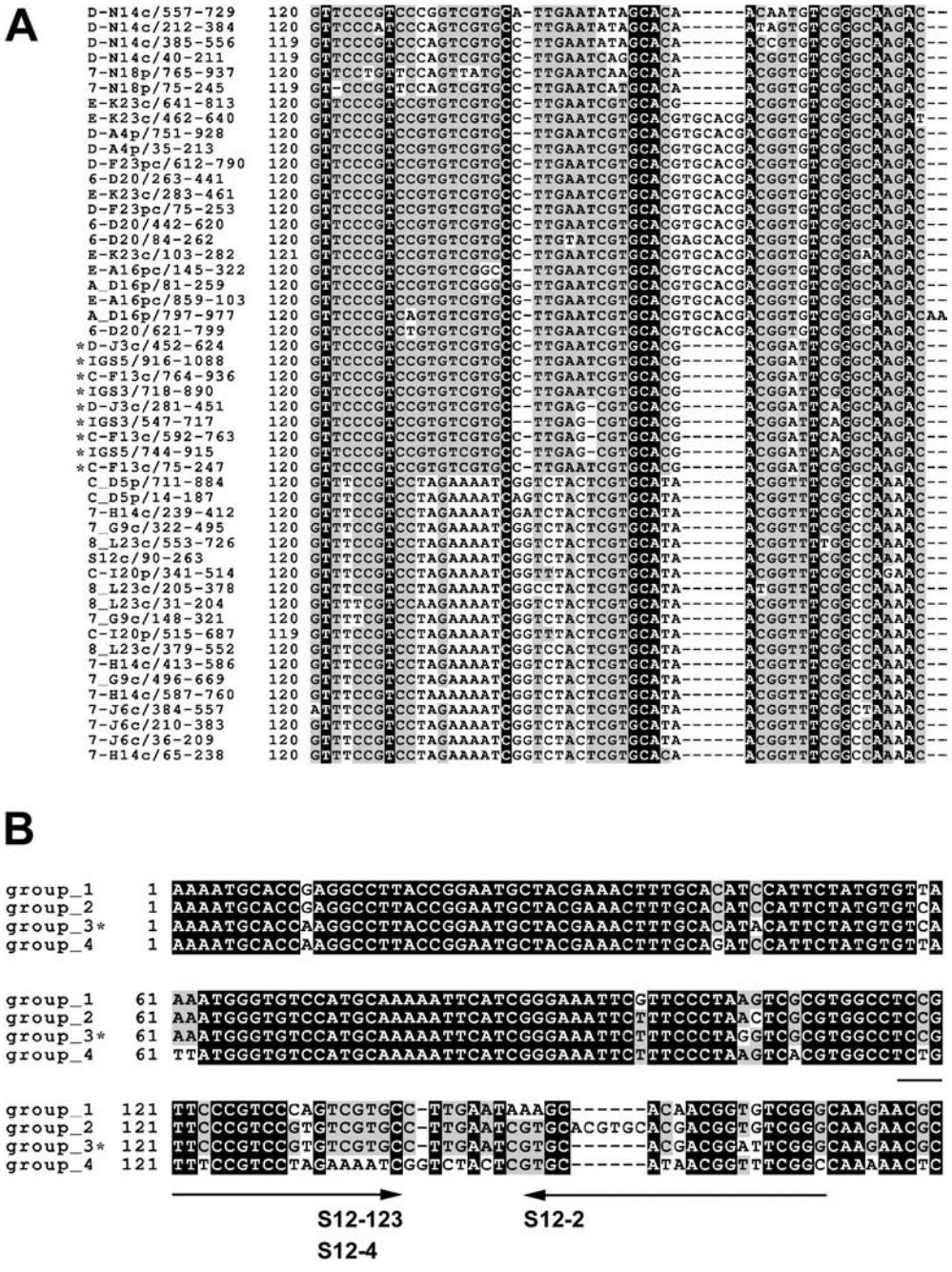


Fig. 2A, B Multiple sequence alignment of the IGS A subrepeats and S12 satellite repeats of *V. sativa*. **A** Segment of the alignment of all available sequences showing the variable 3′ ends of the monomers. Completely or partially conserved nucleotides are indicated by *black* and *gray* backgrounds, respectively. The monomers are labeled with the clone name followed by the position of the monomer within the clone. The clones marked with an *asterisk* represent IGS A subrepeats obtained from either the two full-length IGS clones (VS-IGS-3 and -5) or from partial IGS sequences retrieved from the shotgun genomic DNA library (clones C-F13c and D-J3c). Clone S12 corresponds to the sequence isolated by Nouzová et al. (2001). All other monomers were extracted from the clones of the IGS-like S12 satellite repeat isolated from the shotgun DNA library. **B** Comparison of consensus sequences representing the four groups of monomers. The groups were defined based on the multiple sequence alignment of all sequences and consisted of 6 (group 1), 16 (group 2), 9 (group 3), and 19 (group 4) monomers, respectively (see Electronic Supplementary Material for monomer alignments within individual groups). Group 3, which includes only IGS A subrepeats, is marked with an *asterisk*. The positions and orientations of primers used for primed in situ DNA labeling (PRINS) are indicated with *arrows* and the primer sequences are given in Materials and methods

oligonucleotide primers. Three primers were designed, based on the multiple alignment of the monomers so that their 3′ ends contained sequences specific for one or more repeat subfamilies (Fig. 2B). As the homology of the primer 3′ end to the template is crucial for its successful extension by a Taq polymerase, this arrangement should allow selective labeling of the corresponding sequence variants on chromosomes using PRINS. Two primers (S12-4 and S12-123) were designed to discriminate group 4 variants from the sequence common to the other three groups, including IGS subrepeats. The PRINS reaction with the primer S12-123 resulted in labeling of bands on chromosomes 4, 5 and 6 as well as of both nucleolus organizing regions containing the rDNA genes on chromosomes 2 and 3 (Fig. 3A). These signals thus corresponded to all genomic loci containing IGS A subrepeats or IGS-like satellites as revealed by fluorescence in situ hybridization (FISH) with the S12 probe (Nouzová et al. 2001). On the other hand, the reaction using the group 4-specific primer S12-4 generated signals only at chromosomes 4 and 5, suggesting that this sequence subfamily occurs at only these two genomic locations (Fig. 3B). The third primer (S12-2) was targeted to a 6 bp sequence duplication specific for group 2 and it selectively labeled only bands on chromosomes 5 and 6 (Fig. 3C). In order to confirm PRINS results using an alternative technique, we

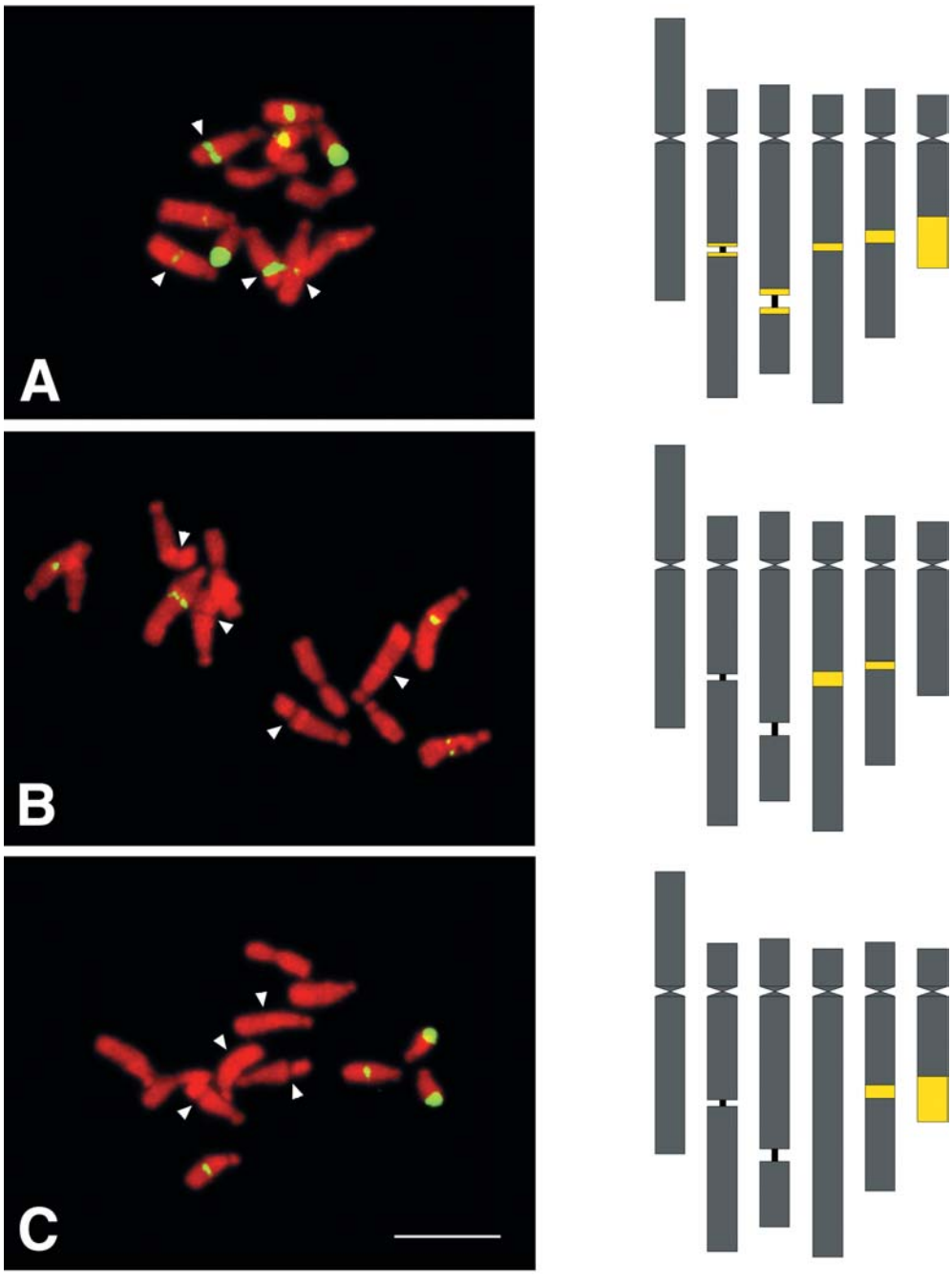


Fig. 3 Localization of the repeat subfamilies on *V. sativa* ($2n$ =12) mitotic chromosomes using PRINS with primer S12-123 (**A**), S12-4 (**B**), and S12-2 (**C**). The PRINS signals are *yellow-green*; chromosomes were counterstained with 4′,6-diamino-2-phenylindole, which is displayed in *red* for better contrast with the PRINS signals. The positions of nucleolus organizing regions on chromosomes 2 and 3 are marked with *arrowheads*. *Bar* represents 10 μm. Signal positions are also schematically depicted on the *right* of each panel

performed FISH using two oligonucleotide probes discriminating the group 4 subfamily from the remaining sequences. These hybridizations produced the same labeling patterns as PRINS with the primers S12-4 and S12-123 (data not shown). Thus, the PRINS and FISH experiments showed that the sequence subfamilies of the S12 repeats are not homogeneously dispersed in all regions containing IGS-like sequences but instead at least some of them are selectively amplified at the limited number of these loci.

Discussion

The occurrence of independent satellite repeats homologous to the IGS subrepeats in *V. sativa* and other legume species (Maggini et al. 1991; Unfried et al. 1991; Falquet et al. 1997) raises questions about the relations of these repeats to IGS and the mechanisms of their origin and amplification in the genome. A comparison of available sequences revealed that the satellite repeats are always more similar to the corresponding IGS subrepeats from the same species than to the satellite repeats or IGS sequences from other plants (data not shown). This can be

explained by the supposed transposition of the satellite repeats to IGS in these species (Maggini et al. 1991), as well as by the existence of the reverse mechanism, causing the spreading of sequences from the IGS to genomic locations outside the rDNA clusters (De Lucchini et al. 1988; Unfried et al. 1991). Although definite conclusions cannot be drawn owing to insufficient data, there are several findings that appear to be in favor of the latter hypothesis: (1) The IGS-like satellite repeats are present only in the subset of related species sharing similar spacer sequences, as was shown for *Vigna* (Unfried et al. 1991) and *Vicia* (Nouzová et al. 2001). (2) Although the IGSs of *Vigna, Phaseolus* and *Vicia* contain at least two different subrepeats, the IGS-like satellites are in all cases homologous to the subrepeat closest to the 3′ end of the 25S rRNA genes. This position may thus be important for molecular mechanisms causing transposition of the repeats to other genomic locations. In *V. radiata* the termination of transcription has been mapped to this region (Schiebel et al. 1989), which inspired the proposition of a reverse transcription and reintegration-based origin for IGS-like satellite repeats (Unfried et al. 1991). (3) Several repeats showing homology to IGS but not associated with rDNA have also been reported from yeast, animals and insects. In addition to satellite repeats with short monomers of several hundreds of base pairs (De Lucchini et al. 1988; Lohe and Roberts 1990), larger segments (up to 13.5 kb) of rDNA spacers have also been detected outside rDNA loci in mouse (Kominami and Muramatsu 1987) and recently also in the plant *Solanum bulbocastanum* (Stupar et al. 2002). It is quite likely that such sequences, spanning significant portions of the ribosomal spacers and displaying their complex structure, originated from IGS and not vice versa. On the other hand, it is not clear whether they were transposed by the same mechanisms as the IGS-like satellites with the short monomers.

The identification of sequence subfamilies of the IGS-like S12 repeat and their successful visualization using PRINS on mitotic chromosomes described here provided an additional tool for investigation of the evolutionary dynamics of these satellite repeats. Discrimination of the variants showing minor sequence differences from the consensus revealed that these subfamilies are amplified only at the subset of loci containing S12 repeats and are not detectable at the loci containing rDNA (Fig. 3). As the predominant sequence variant, which is also present in the rDNA spacers, occurs in all S12 loci, these findings suggest that the sequence subfamilies corresponding to groups 2 and 4 (Fig. 2) evolved after the spreading of the S12 repeats in the genome and further support the hypothesis of their origin from the rDNA spacer.

Acknowledgements We thank Ms. H. Štěpančiková and Ms. O. Šonková for excellent technical assistance, and Ms. S. Rafelski for assistance in preparation of the manuscript. This work was supported by grants AVOZ 5051902 and GA CR 204/04/1207. The experiments have been carried out in compliance with the current laws governing genetic experimentation in the Czech Republic.

References

Altschul SF, Madden TL, Schäffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res 25:3389–3402

Benson G (1999) Tandem repeats finder: a program to analyze DNA sequences. Nucleic Acids Res 27:573–580

Borisjuk N, Borisjuk L, Komarnytsky S, Timeva S, Hemleben V, Gleba Y, Raskin I (2000) Tobacco ribosomal DNA spacer element stimulates amplification and expression of heterologous genes. Nature Biotech 18:1303–1306

De Lucchini S, Andronico F, Andreazzoli M, Giuliani M, Savino R, Nardi I (1988) Extra-ribosomal spacer sequences in *Triturus*. J Mol Biol 204:805–813

Dellaporta SL, Wood J, Hicks JB (1983) A plant DNA minipreparation: version II. Plant Mol Biol Rep 1:19–21

Falquet J, Creusot F, Dron M (1997) Molecular analysis of *Phaseolus vulgaris* rDNA unit and characterization of a satellite DNA homologous to IGS subrepeats. Plant Physiol Biochem 35:611–622

Hernandez P, Martin-Parras L, Martinez-Robles ML, Schvartzman JB (1993) Conserved features in the mode of replication of eukaryotic ribosomal RNA genes. EMBO J 12:1475–1485

Kato A, Yakura K, Tanifuji S (1985) Repeated DNA sequences found in the large spacer of *Vicia faba* rDNA. Biochim Biophys Acta 825:411–415

Kato A, Nakajima T, Yamashita J, Yakura K, Tanifuji S (1990) The structure of the large spacer region of the rDNA in *Vicia faba* and *Pisum sativum*. Plant Mol Biol 14:983–993

Kominami R, Muramatsu M (1987) Amplified ribosomal spacer sequence: structure and evolutionary origin. J Mol Biol 193:217–222

Lohe AR, Roberts PA (1990) An unusual Y chromosome of *Drosophila simulans* carrying amplified rDNA spacer without rRNA genes. Genetics 125:399–406

Macas J, Požárková D, Navrátilová A, Nouzová M, Neumann P (2000) Two new families of tandem repeats isolated from genus *Vicia* using genomic self-priming PCR. Mol Gen Genet 263:741–751

Macas J, Mészáros T, Nouzová M (2002) PlantSat: a specialized database for plant satellite repeats. Bioinformatics 18:28–35

Maggini F, Cremonini R, Zolfino C, Tucci GF, D'Ovidio R, Delre V, De Pace PG, Scarascia Mugnozza GT, Cionini PG (1991) Structure and chromosomal localization of DNA sequences related to ribosomal subrepeats in *Vicia faba*. Chromosoma 100:229–234

Nouzová M, Neumann P, Navrátilová A, Galbraith DW, Macas J (2001) Microarray-based survey of repetitive genomic sequences in *Vicia* spp. Plant Mol Biol 45:229–244

Pearson WR, Lipman DJ (1988) Improved tools for biological sequence comparison. Proc Natl Acad Sci U S A 85:2444–2448

Piller KJ, Baerson SR, Polans NO, Kaufman LS (1990) Structural analysis of the short length ribosomal DNA variant from *Pisum sativum* L. cv. Alaska. Nucleic Acids Res 18:3135–3145

Rogers SO, Bendich AJ (1987) Ribosomal RNA genes in plants: variability in copy number and in the intergenic spacer. Plant Mol Biol 9:509–520

Sanger F, Nicklen D, Coulson AR (1977) DNA sequencing with chain terminating inhibitors. Proc Natl Acad Sci U S A 74:5463–5467

Schiebel K, von Waldburg G, Gerstner J, Hemleben V (1989) Termination of transcription of ribosomal RNA genes of mung bean occurs within a 175 bp repetitive element of the spacer region. Mol Gen Genet 218:302–307

Sonnhammer ELL, Durbin R (1995) A dot-matrix program with dynamic threshold control suited for genomic DNA and protein sequence analysis. Gene 167:GC1–10

Staden R (1996) The Staden sequence analysis package. Mol Biotechnol 5:233–241

Stupar RM, Song JQ, Tek AL, Cheng ZK, Dong FG, Jiang JM (2002) Highly condensed potato pericentromeric heterochro-

matin contains rDNA-related tandem repeats. Genetics 162:1435–1444

Suzuki A, Tanifuji S, Komeda Y, Kato A (1996) Structural and functional characterization of the intergenic spacer region of the rDNA in *Daucus carota*. Plant Cell Physiol 37:233–238

Takaiwa F, Kikuchi S, Oono K (1990) The complete nucleotide sequence of the intergenic spacer between 25S and 17S rDNAs in rice. Plant Mol Biol 15:933–935

Ueki M, Uchizawa E, Yakura K (1992) The nucleotide sequence of the rDNA intergenic spacer region in a wild species of the genus *Vicia, V. angustifolia*. Plant Mol Biol 18:175–178

Unfried K, Schiebel K, Hemleben V (1991) Subrepeats of rDNA intergenic spacer present as prominent independent satellite DNA in *Vigna radiata* but not in *Vigna angularis*. Gene 99:63–68

Urawa H, Hidaka M, Ishiguro S, Okada K, Horiuchi T (2001) Enhanced homologous recombination caused by the non-transcribed spacer of the rDNA in *Arabidopsis*. Mol Genet Genomics 266:546–555